RFID BASED DOOR ACCESS SYSTEM

(Computer Networking Project – Cisco Packet Tracer)

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1. Introduction

The RFID Based Door Access System is a smart security project developed in Cisco Packet Tracer.
This project uses RFID technology and IoT devices to control door access automatically. Only authorized users with a valid RFID card can unlock the door, while unauthorized access attempts keep the door locked.
The system demonstrates the practical implementation of:

- Computer Networking
- IoT Automation
- RFID Security
- DHCP Configuration
- Smart Access Control Systems

2. Abstract

The RFID Based Door Access System is designed to provide secure and automated door access using RFID technology in a network environment. The project is implemented in Cisco Packet Tracer using IoT devices such as an RFID Reader, RFID Card, Smart Door, Router, Switches, and IoT Server.

In this system, the RFID Reader scans the RFID card and sends the card information to the IoT server through the network. The server checks whether the card ID is valid or invalid based on predefined automation rules. If the card is authorized, the smart door unlocks automatically; otherwise, the door remains locked.

The project improves security, reduces manual effort, and demonstrates the integration of networking with IoT automation systems.

3. Project Objectives

- To design a secure smart door access system using RFID technology.
- To understand the implementation of IoT devices in networking.
- To configure routers, switches, and servers in Cisco Packet Tracer.
- To automate door locking and unlocking operations.
- To improve security by allowing only authorized access.
- To understand DHCP and IP addressing concepts.
- To demonstrate real-world applications of smart security systems.

4. Problem Statement

Traditional lock systems require physical keys, which can be lost, duplicated, or stolen. Manual security systems are less efficient and require human monitoring.

This project solves these issues by implementing:

- Automatic authentication
- Smart door control
- RFID-based verification
- Network-based monitoring

5. Scope of the Project

This project can be used in:

- Offices
- Universities
- Smart homes
- Laboratories
- Banks
- Restricted security areas

Future enhancements may include:

- Fingerprint authentication
- Face recognition
- Mobile app integration
- Cloud monitoring

6. Software and Devices Used

Software

- Cisco Packet Tracer

Hardware/Network Components

- Router – Provides network connectivity
- Switches – Connect network devices
- Server – Stores IoT automation rules
- RFID Reader – Reads RFID card data
- RFID Card – Used for authentication
- Smart Door – Locks/unlocks automatically

7. Network Topology Description

The network topology consists of:

- One router connected to the main switch
- Two switches for device connectivity
- One IoT server
- One RFID Reader
- One RFID Card
- One smart door

The RFID Reader sends card information to the IoT Server through the switches and router. The server processes the information and controls the smart door based on configured rules.

8. IP Addressing Table

Device	IP Address
Router (R1)	192.168.1.1
IoT Server	192.168.1.51
Network	192.168.1.0/24

9. Router Configuration

The router is configured with IP address, DHCP service, and network gateway.

Example configuration:

- enable
- configure terminal
- hostname R1
- interface gigabitEthernet 0/0
- ip address 192.168.1.1 255.255.255.0
- no shutdown
- exit
- ip dhcp pool iot
- network 192.168.1.0 255.255.255.0
- default-router 192.168.1.1

10. IoT Server Configuration

Setting	Value
Server IP	192.168.1.51
Default Gateway	192.168.1.1
Username/Password	admin / admin

The IoT server is responsible for:

- Managing IoT devices
- Processing RFID authentication
- Executing automation rules

11. Automation Rules

- Rule 1: If Card ID = 1001 → Valid
- Rule 2: If Card ID ≠ 1001 → Invalid
- Rule 3: If Valid → Door Unlock
- Rule 4: If Invalid → Door Lock

12. Working of the Project

Step 1: RFID card is scanned by RFID Reader.
Step 2: Data is sent to IoT Server.
Step 3: Server verifies card ID.
Step 4: If valid → Door unlocks.
Step 5: If invalid → Door remains locked.

13. Advantages of the System

- Improved security
- Automated access control
- Fast authentication
- Easy monitoring
- Reduced manual work
- Low-cost smart security system

14. Limitations

- Depends on network connectivity
- RFID cards can be damaged
- Limited predefined IDs
- Simulation-based system

15. Future Improvements

- Biometric verification
- Mobile notifications

- Cloud database integration
- CCTV monitoring
- AI-based security analysis

16. Conclusion

The RFID Based Door Access System successfully demonstrates the integration of computer networking and IoT automation using Cisco Packet Tracer. The system provides secure and automatic door access through RFID authentication.

It helps students understand:

- Networking concepts
- IoT automation
- Smart security systems
- Real-world RFID implementation

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